

Database Technologies – Assignment 8 Use sub-query to solve following problems. Note: To solve below queries, use “sales” database

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-98849-sakshi@localhost> SELECT *FROM orders
-> WHERE cnum=(SELECT cnum from customers WHERE cname='Cisneros');

+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.015 sec)
```

7. Extract all the orders of Motika.

```
KD3-98849-sakshi@localhost> SELECT * FROM orders
-> WHERE snum=(SELECT snum FROM salespeople WHERE sname='motika');

+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.015 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD3-98849-sakshi@localhost> SELECT * FROM orders
-> WHERE cnum IN(SELECT cnum FROM customers WHERE city='london');

+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.014 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3-98849-sakshi@localhost> SELECT *FROM orders
-> WHERE amt >ALL(SELECT snum FROM customers WHERE city='LONDON');
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.014 sec)

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3-98849-sakshi@localhost> SELECT * FROM customers
-> WHERE rating >=ANY(SELECT rating FROM customers c
-> INNER JOIN salespeople s ON s.snum=c.snum
-> WHERE sname='Serres');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.034 sec)

11. Select customers who have a greater rating than any other customer in Rome.

```
KD3-98849-sakshi@localhost> SELECT *FROM customers
-> WHERE rating>ANY(SELECT rating FROM customers WHERE city='rome
');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.012 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD3-98849-sakshi@localhost> SELECT * FROM orders
-> WHERE amt > ANY(SELECT amt FROM orders WHERE odate='1990-10-04');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.016 sec)

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-98849-sakshi@localhost> select * from orders
-> WHERE amt < ANY(SELECT amt FROM orders o
-> INNER JOIN customers c ON o.cnum=c.cnum
-> WHERE city='San jose');
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002

8 rows in set (0.014 sec)

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3-98849-sakshi@localhost> SELECT o.amt,c.rating,c.cname
-> FROM orders o
-> JOIN customers c ON o.cnum=c.cnum
-> WHERE amt>(select AVG(amt)FROM orders );
```

amt	rating	cname
5160.45	200	Liu
9891.88	100	Clemens
4723.00	100	Clemens

3 rows in set (0.013 sec)

14. Select those customers whose rating are higher than every customer in London.

```
KD3-98849-sakshi@localhost> SELECT *FROM customers
-> WHERE rating>ANY(SELECT rating FROM customers WHERE city='rome'
);
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.012 sec)

5. Write a query that will find all salespeople who have no customers located in their city.
(Hint: Choose the correct keyword ANY/ALL).

```
KD3-98849-sakshi@localhost> SELECT sname,snum,city FROM salespeople s
-> WHERE s.city != ALL(
-> SELECT c.city
-> FROM customers c
-> WHERE c.snum=s.snum
-> );
```

sname	snum	city
Motika	1004	London
Rifkin	1007	Barcelona
Axelrod	1003	New York

3 rows in set (0.012 sec)

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3-98849-sakshi@localhost> SELECT s.snum, s.sname
-> FROM salespeople s
-> JOIN customers c
-> ON s.snum = c.snum
-> GROUP BY s.snum, s.sname
-> HAVING COUNT(c.cnum) > 1;
```

```
+-----+-----+
```

```
| snum | sname |
```

```
+-----+-----+
```

```
| 1001 | Peel |
```

```
| 1002 | Serres |
```

```
+-----+-----+
```

```
2 rows in set (0.257 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3-98849-sakshi@localhost> SELECT s.snum, s.sname, s.city, COUNT(c.cnum) AS customer_count
-> FROM salespeople s
-> JOIN customers c
-> ON s.snum = c.snum
-> GROUP BY s.snum, s.sname, s.city
-> HAVING COUNT(c.cnum) > 1;
```

```
+-----+-----+-----+-----+
```

```
| snum | sname | city | customer_count |
```

```
+-----+-----+-----+-----+
```

```
| 1001 | Peel | London | 2 |
```

```
| 1002 | Serres | San Jose | 2 |
```

```
+-----+-----+-----+-----+
```

```
2 rows in set (0.034 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD3-98849-sakshi@localhost> SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (SELECT MAX(amt) FROM orders);
```

```
+-----+-----+
```

```
| snum | total_amount |
```

```
+-----+-----+
```

```
| 1001 | 15382.07 |
```

```
+-----+-----+
```

```
1 row in set (0.092 sec)
```